

The maximum size of an almost-equidistant set in $\mathbb{R}^5$ is sixteen

A computer-assisted determination of $f(5)$

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Background

A finite set of points in $\mathbb{R}^d$ is *almost equidistant* if among any three of its points, some two are at unit distance. Let $f(d)$ denote the largest size of an almost-equidistant set in $\mathbb{R}^d$. Classically $f(2) = 7$ (the Moser spindle) and $f(3) = 10$ (Györey [4]); Balko, Pór, Scheucher, Swanepoel and Valtr [1] re-proved these with computer assistance and established $12 \leq f(4) \leq 13$ as well as

$$16 \leq f(5) \leq 20,$$

the lower bound being the Larman–Rogers half-cube construction [2]. The value $f(4) = 12$ (Conjecture 1 of [1]) was resolved in the accompanying note `f4_equals_12.pdf` of this repository. No improvement on $f(5)$ appeared in the literature since [1]; the best asymptotic bound is $f(d) = O(d^{4/3})$ [3].

Result. $f(5) = 16$ (computer-assisted proof; see the caveats below). Moreover, no almost-equidistant set of 17, 18, 19 or 20 points exists in $\mathbb{R}^5$, each level being eliminated independently by certified interval computation.

Reduction

Following [1], the unit-distance graph G of an almost-equidistant set in $\mathbb{R}^5$ satisfies: (i) $\alpha(G) \leq 2$ (the complement is triangle-free); (ii) no K_7 (at most $d+1$ points of $\mathbb{R}^d$ are pairwise at unit distance); (iii) no $K_{3,3,3}$ subgraph (Lemma 11 of [1]: three 3-point classes with all cross distances 1 would span three pairwise orthogonal ≥ 2 -dimensional affine hulls, hence dimension ≥ 6). Deleting removable edges gives a spanning *minimal* such graph (complement a *maximal* triangle-free graph, still satisfying (ii) and (iii)), and any realization — n distinct points with all edges at unit distance, non-edges unconstrained — restricts to it. Hence if no minimal abstract almost-equidistant graph on n vertices is realizable with n distinct points in $\mathbb{R}^5$, then $f(5) < n$.

The minimal graphs were generated from the maximal triangle-free graphs (program `triangleramsey` [5, 6]) by filtering with independently written checkers; the counts match [1] (Table 2, $d = 5$) at every order:

n	17	18	19	20	21
maximal triangle-free graphs	164 796	1 337 848	13 734 745	178 587 364	2 911 304 940
minimal abstract a.e. graphs	12 654	8 825	340	8	0

(The $n = 21$ column re-verifies $f(5) \leq 20$ of [1] independently.) As an independent check of the class definition and the filters, a from-scratch vertex-growth enumerator reproduces the $d = 5$ column of Table 3 of [1] at every order it reaches (7, 14, 38, 106, 402, 1817, 11 132, 86 053, 803 299, 7 623 096 for $n = 4, \dots, 13$) and the minimal counts up to $n = 13$; at $n = 13$ the 242 minimal graphs agree *as sets* between the two routes, matched one-to-one up to isomorphism. The lower bound $f(5) \geq 16$ is verified in integer arithmetic: the odd half-cube $\{v \in \{\pm 1\}^5 : v \text{ odd } \#(+1)\text{s}\}/\sqrt{8}$ is a two-distance set (squared distances 1 and 2) whose non-unit graph is the triangle-free Clebsch graph.

Method

The decision engine (`ckernel15.c`) is the $\mathbb{R}^5$ port of the certified branch-and-prune engine that proved $f(4) = 12$ (Appendix B of the $f(4)$ note), with only the dimension-forced changes: rigorous outward-rounded interval arithmetic over IEEE doubles (cos/sin padded by 8 ulp); exact K_6 (unit 5-simplex) or K_5 seeds; binary sphere placements from ≥ 5 placed neighbours (5×5 interval Gauss–Jordan, Krawczyk contraction on the best-conditioned 5-subset, surplus neighbours as certified constraints, Gauss–Seidel re-contraction sweeps); circle placements from exactly 4 placed neighbours (circumcentre by a 3×3 interval Gram system, orthonormal complement basis, adaptive θ -subdivision with a runtime canary); merged-root tangencies resolved by one-dimensional segment subdivision; and two certified coincidence rules that discard branches on which two non-adjacent vertices are forced to coincide (root separation over 5-subsets of common placed neighbours; the bisector-hyperplane rule with 6 common placed neighbours of certifiably nonzero affine volume). A branch is discarded only if an interval certifiably excludes a required value or coincidence is forced; eliminating every branch of one elimination order certifies that the graph has no realization with all points distinct.

Two search-management additions (soundness-neutral) were needed beyond the $f(4)$ engine, both prompted by the hardest $\sim 0.3\%$ of the 17-vertex graphs: (i) a *per-cell node quota* — a circle cell whose subtree exceeds a fixed node budget is abandoned and split, since a cell can be arbitrarily expensive to kill outright even when both its halves die almost immediately (the $f(4)$ -era engine split cells only when their subtrees *parked* boxes, which such cells never do); and (ii) a certified *mean-value enclosure* for the circle parametrization (intersected with the direct enclosure), quadratically tighter on narrow cells.

Orchestration. Elimination orders are generated circle-late (and, for stubborn graphs, variants with a deliberately early circle stage, whose parameter can be tiled into slices); the driver races candidate orders and certifies with whichever first has its entire tiling killed. Almost all searches are circle-free binary trees (12 528 of 12 654 at $n = 17$; 8 813 of 8 825 at $n = 18$; 339 of 340; 7 of 8), and every candidate graph admits a valid order — no two-parameter stage is ever required. For the final 26 graphs the driver enumerated *all* (seed clique, first-circle-vertex) parametrizations (~ 20 – 30 per graph): for every one of the 26, at least one parametrization yields a 192-slice tiling in which every slice is certifiably killed — the graph-specific “hot arcs” of any single parametrization are artifacts of that parametrization, never of the graph.

Controls. The engine kills K_7 (immediately: the seed’s seventh vertex has an empty sphere system) and the 11-vertex graph $K_1 + K_{2,2,2,2,2}$ (an apex adjacent to all vertices of the 5-dimensional cross-polytope graph would have to be its centre, at distance $1/\sqrt{2} \neq 1$; killed in two search nodes); it does *not* kill the realizable controls K_6 , $K_{2,2,2,2,2}$ (the unit cross-polytope) and the 16-vertex half-cube graph of the extremal set — for these, sliced tilings report floor-width surviving cells around the true realizations.

Numerical corroboration. Independent Levenberg–Marquardt optimization (300 restarts per graph, greedy clique-seeded and random initializations) over all 21 827 candidate graphs finds no graph remotely close to realizable: the smallest residual with distinct points is 0.409 (at $n = 17$), versus 10^{-30} reached on realizable controls. No conclusion rests on the numerics.

Execution

All four levels were eliminated on a single desktop (Ryzen 9950X3D, LLVM clang build of the kernel), racing decompositions across all cores:

- $n = 20$: all 8 graphs certified in ~ 5 seconds wall time;
- $n = 19$: all 340 graphs in ~ 3 minutes;

- $n = 18$: all 8 825 graphs in under a minute (8 813 circle-free searches bulk-certified in ~ 10 seconds);
- $n = 17$: all 12 654 graphs — 12 518 by circle-free searches in six seconds of wall time; 110 more by the racing/slicing driver within seconds each; and the final 26 (one structural family: 90–95 edges, degrees 10–12, few K_6 ’s, near-5-regular maximal-triangle-free complements) certified in 670 seconds total by exhausting parametrizations under the final kernel, each via a fully killed 192-slice tiling.

The certificates for these last graphs record the winning parametrization in `hotmap_results.json`; per-graph verdicts for the whole level are in `results_n17.json`. (Development history, including the search-management defects found and fixed along the way, is preserved in `STATUS.md` and the repository log.) Zero surviving branches were reported anywhere; per-graph verdicts and winning decomposition indices are in `results_n17..20.json`. Together with $f(5) \geq 16$: $f(5) = 16$.

En passant, the independent re-derivation of the candidate enumeration uncovered an erratum in Table 3 of [1] (the $d=6$ column omits the $K_{1,3,3,3}$ exclusion for $n \geq 11$; corrected counts and three independent verification routes are documented in `erratum_bpssv_table3_d6.md`; their Table 2 and all theorems are unaffected).

Caveats and reproducibility

The proof is computer-assisted. It assumes IEEE-754 correctly rounded arithmetic, library `cos / sin` accurate to 8 ulp (padded), the engine logic inherited from the $f(4)$ note’s Appendix B with the $\mathbb{R}^5$ modifications listed above, and the completeness of the `triangleramsey` generation [5], cross-validated here against [1] at $n = 17, \dots, 21$ and against a fully independent enumeration up to $n = 13$ (all counts; sets at $n = 13$). Everything needed to reproduce the computation is in this repository: `reproduce5.py -controls then -level 20/19/18/17 (-bulk)`, with `hotmap.py` for the hardest graphs; the kernel compiles automatically. Independent replication — ideally with an independently implemented engine, e.g. one using dyadic fixed-point arithmetic to re-check the recorded certificate tilings — would be very welcome before the result is treated as established.

With $f(2) = 7$, $f(3) = 10$, $f(4) = 12$ and now $f(5) = 16$, the first five values of f are settled; the next open case is $18 \leq f(6) \leq 26$.

References

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